



# *Geometric Signatures of Compositionality Across a Language Model's Lifetime*

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# Introduction

- Inherent Dimension
- Degree of compositionality
- Geometric Signatures
  - Nonlinear dimension
    - intrinsic dimension
  - Linear dimension
- Meaningful compositionality
- Superficial input complexity

# Setup

- Models

- Llama-3-8B
- Mistral-v0.1-7B
- Pythia models (intermediate checkpoints)
  - 14m
  - 70m
  - 160m
  - 410m
  - 1.4b
  - 6.9b
  - 12b

- Datasets

- The Pile
- Controlled grammar
  - Controlling compositionality
    - K-Grams
  - Shuffling

The [quality<sub>1</sub>.ADJ] [nationality<sub>1</sub>.ADJ]  
[job<sub>1</sub>.N] [action<sub>1</sub>.V] the [size<sub>1</sub>.ADJ]  
[texture.ADJ] [color.ADJ] [animal.N] then  
[action<sub>2</sub>.V] the [size<sub>2</sub>.ADJ] [quality<sub>2</sub>.ADJ]  
[nationality<sub>2</sub>.ADJ] [job<sub>2</sub>.N].

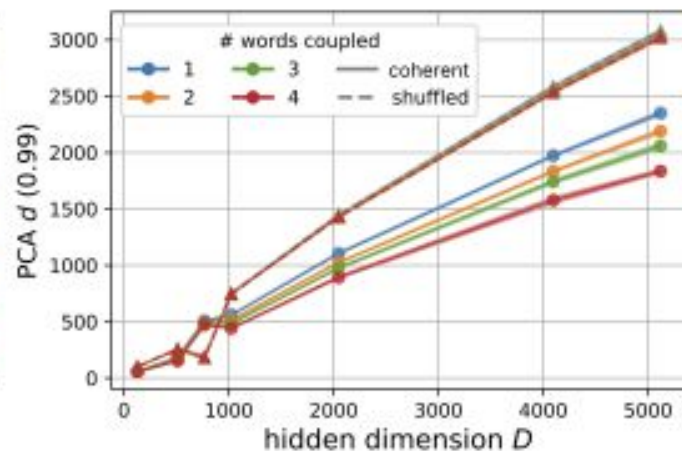
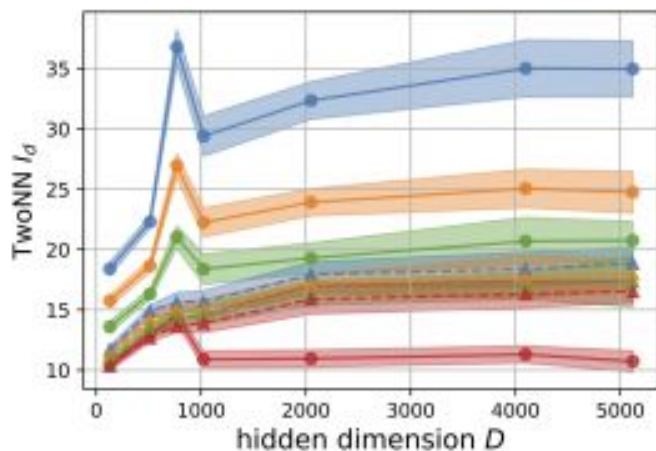
# Measures

- Measuring compositionality
  - Kolmogorov complexity
    - The length of the shortest program in bits needed to generate the dataset.
- Measuring feature complexity
  - Intrinsic dimension
    - TwoNN estimator
  - Linear effective dimension
    - PCA
    - variance cutoff of 99%.

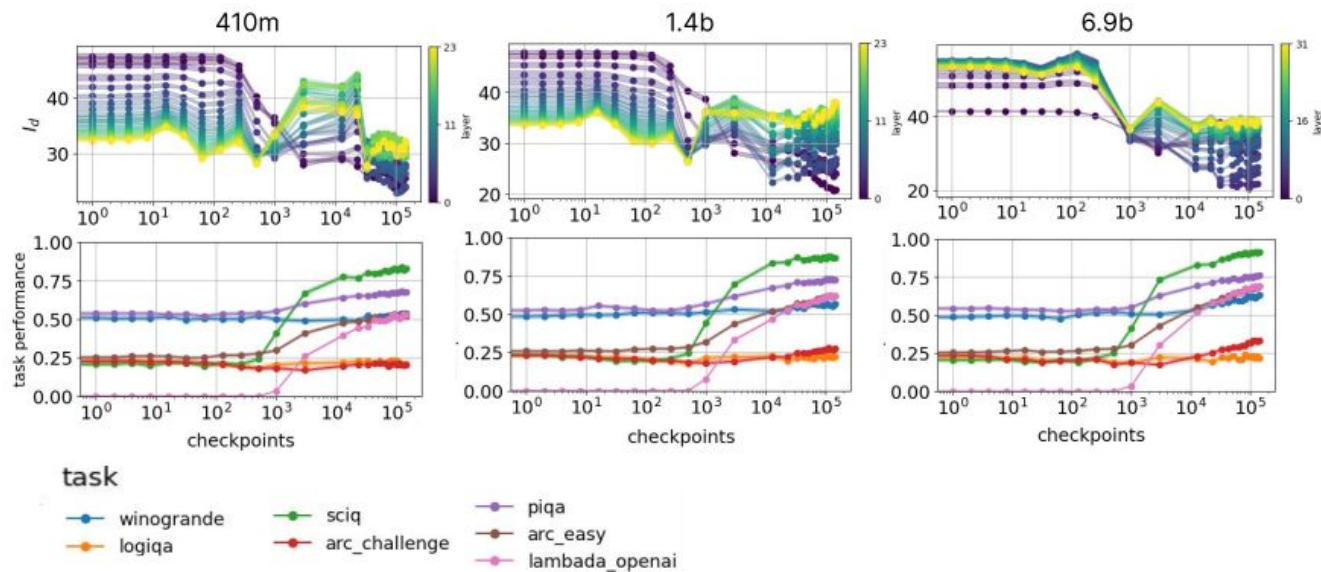


# *Results*

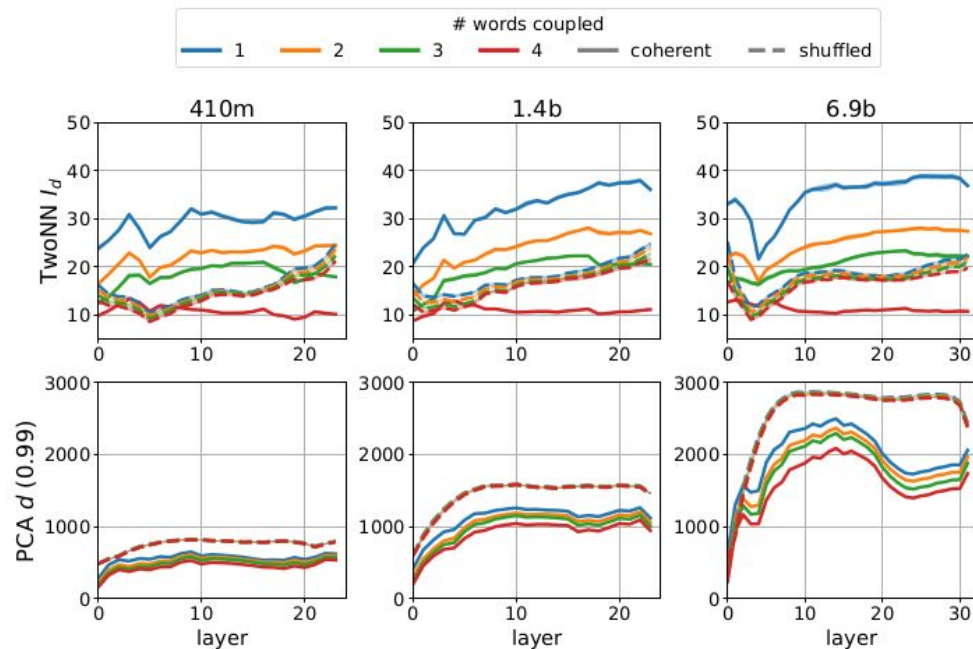
## Nonlinear and linear feature complexity scale differently with model size



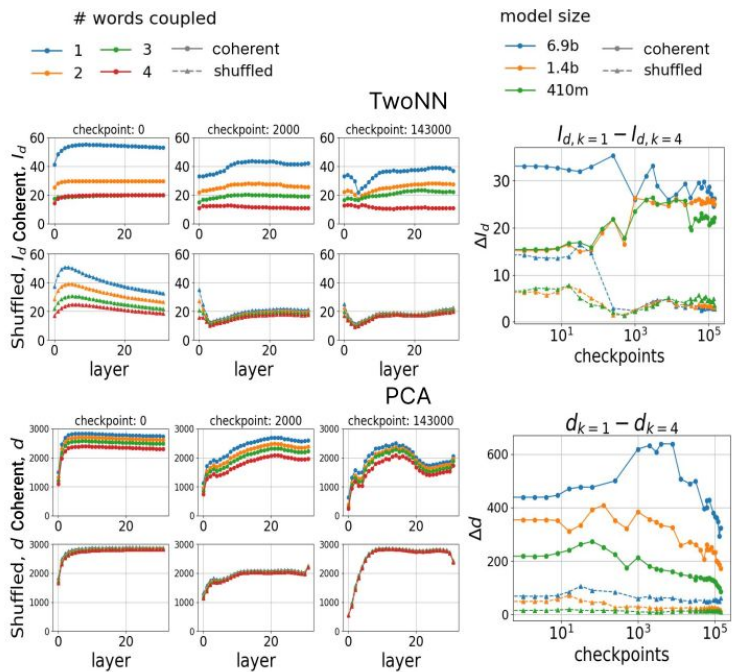
# Feature complexity tracks emergent linguistic abilities over training



# Compositional complexity varying k



# Breaking compositional structure



# Form-meaning dichotomy in representation learning

| Spearman $\rho$ | 14m   | 70m               | 160m               | 410m  | 1.4b  | 6.9b  | 12b   | Llama | Mistral |
|-----------------|-------|-------------------|--------------------|-------|-------|-------|-------|-------|---------|
| $I_d$           | -0.20 | -0.06             | -0.20              | -0.05 | 0.04  | 0.01  | 0.05  | -0.36 | 0.00    |
| $d$             | 0.90* | 0.47 <sup>†</sup> | -0.50 <sup>†</sup> | 0.96* | 0.96* | 0.92* | 0.86* | 1.0*  | 1.0*    |

Table 1: **Spearman correlations between dimensionality and estimated Kolmogorov complexity.** Spearman correlations  $\rho$  between the gzipped dataset size (KB) and representational dimensionality (rows), averaged over layers, for all **Pythia** sizes (left 7 columns) and **Llama** and **Mistral** (right 2 columns). Values with \* are significant at a p-value threshold of 0.05, and <sup>†</sup> at 0.1. Across LMs, average-layer  $I_d$  is not correlated to the estimated KC, or surface complexity, of datasets. Linear  $d$  is highly correlated to estimated KC, except the outlier 160m.

This suggests surface complexity, already present in the LM’s inputs, is preserved by the architecture, while meaning complexity is instead learned over training.



# Any Question?

*Thank you for your attention!*



# Reference

- Jin Hwa Lee, Thomas Jiralerspong, Lei Yu, Yoshua Bengio, Emily Cheng. Geometric Signatures of Compositionality Across a Language Model's Lifetime (Under review at ARR)